

A10_CAT-016

 **Video Explanation:**  **Assignment10.mp4**

MediCare Connect – Hospital Appointment & Healthcare Management System

Dear Candidates,

We are pleased to inform you that you have successfully passed the first round of the selection process!! 🎉

Your application and skills have impressed us, and we are excited to move forward with you in the next stages. We are excited to have you explore this opportunity with us. This project is designed to assess your skills, creativity, and problem-solving abilities. It will help us understand how you approach challenges and your ability to deliver high-quality solutions.

Project Overview and Discussion

What is MediCare Connect?

MediCare Connect is a modern healthcare management platform that connects patients with doctors and hospitals through a centralized online system. Patients can book appointments, manage medical records, make payments, and receive healthcare services efficiently.

Doctors can manage schedules, appointments, and patient consultations, while administrators can oversee the entire healthcare ecosystem.

Why Should We Develop This Project?

Traditional appointment systems often involve long waiting times, manual paperwork, and poor communication between patients and healthcare providers.

MediCare Connect aims to:

- Digitize appointment booking.
 - Reduce patient waiting time.
 - Improve doctor schedule management.
 - Maintain healthcare records securely.
 - Provide a seamless healthcare experience.
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Roles:

Patient

- Creates account
- Searches doctors
- Books appointments
- Makes payments
- Views appointment history
- Reviews doctors

Doctor

- Creates professional profile
- Manages schedules
- Accepts or rejects appointments
- Updates consultation status

Admin

- Manages users
- Verifies doctors
- Oversees appointments
- Monitors payments

- Generates reports
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Similar Websites / Design Inspiration

- [Apollo 24/7](#)
 - [Novant Health](#)
 - [UsNews](#)
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Data Design

Collection 1: Users

Fields:

- name
 - email
 - role
 - Photo
 - phone
 - gender
 - createdAt
 - status
-

Collection 2: Doctors

Fields:

- doctorName
- specialization
- qualifications
- experience
- consultationFee

- hospitalName
 - profileImage
 - availableDays
 - availableSlots
 - verificationStatus
-

Collection 3: Appointments

Fields:

- patientId
 - doctorId
 - appointmentDate
 - appointmentTime
 - appointmentStatus
 - symptoms
 - paymentStatus
-

Collection 4: Reviews

Fields:

- patientId
 - doctorId
 - rating
 - reviewText
 - createdAt
-

Collection 5: Payments

Fields:

- appointmentId
- patientId
- doctorId

- amount
 - transactionId
 - paymentDate
-

Collection 6: Prescriptions

Fields:

- doctorId
 - patientId
 - appointmentId
 - diagnosis
 - medications
 - notes
 - createdAt
-

Ensure the Following Things to Get 100% Mark

- Minimum 20 meaningful commits on Client Side.
 - Minimum 12 meaningful commits on Server Side.
 - Meaningful commit messages.
 - Professional README file.
 - Environment variables for Firebase configuration.
 - Environment variables for MongoDB credentials.
 - Responsive design.
 - Proper validation.
 - Secure API routes.
 - JWT implementation.
 - Role-based authorization.
 - No copied design.
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Deployment Guideline

- No CORS issues.
 - No 404 errors.
 - No 504 errors.
 - The server must work properly in production.
 - All private routes must work after page refresh.
 - Live site should remain fully functional.
 - Logged-in users must remain authenticated after reload.
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Layout & Page Structure

Navbar

Must Include:

- Logo
 - Home
 - Find Doctors
 - About Us
 - Contact Us
 - Dashboard
 - Login/Register
 - User Profile Dropdown
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Footer

Must Include:

- Logo
- Quick Links
- Contact Information
- Emergency Hotline
- Social Media Links

- Copyright
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Main Pages

Public Pages

- Home
- Find Doctors
- Doctor Details
- About Us
- Contact Us
- Login
- Register

Private Pages

- Dashboard
 - Profile
-

Authentication System

Register

Fields:

- Name
- Email
- Photo URL/Photo Upload
- Password

Requirements:

- Strong password validation (At least 6 characters including at least one number and one special character)
- Success notification

Login

Requirements:

- Email & Password Login
 - Google Login
 - JWT Authentication
 - Error Handling
-

Home Page

Banner Section

Must Include:

- Healthcare-themed banner
 - Call-to-action button
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Featured Doctors (Dynamic)

Display doctors from the database.

Information:

- Name
 - Specialization
 - Experience
 - Consultation Fee
-

Medical Specializations (Static)

Examples:

- Cardiology
 - Neurology
 - Orthopedics
 - Pediatrics
 - Dermatology
-

Platform Statistics (Dynamic)

Display:

- Total Doctors
 - Total Patients
 - Total Appointments
 - Total Reviews
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Framer Motion Requirement

Implement animation in at least two sections

Extra Section 1 (Dynamic)

Patient Success Stories

Show real patient testimonials. (Hint : you can fetch data from reviews collection)

Extra Section 2 (Static)

Why Choose MediCare Connect

Display healthcare benefits and platform advantages.

Dashboard Layout

Patient Dashboard

Dashboard Overview

Display:

- Upcoming Appointments
- Appointment History
- Total Payments
- Favorite Doctors

MY Profile—

My Appointments

CRUD Features:

- View Appointment
- Reschedule Appointment
- Cancel Appointment

Payment History

Display:

- Paid Appointments
 - Transaction Records
-

My Reviews

CRUD Features:

- Add Review
 - Update Review
 - Delete Review
-

Doctor Dashboard

Dashboard Overview

Display:

- Total Patients
 - Today's Appointments
 - Reviews Received
-

Manage Schedule

CRUD Features:

- Add Schedule
 - Update Schedule
 - Remove Schedule
-

Appointment Requests

Features:

- Accept Appointment
- Reject Appointment

- Mark Completed (After marking as complete, navigate to the prescription management route)
-

Prescription Management

CRUD Features:

- Create Prescription
 - Update Prescription
-

Profile Management

Update:

- Qualifications
 - Experience
 - Consultation Fee
 - Available Slots
-

Admin Dashboard

Dashboard Overview

Manage Users

Features:

- View Users
- Delete User
- Suspend User

Manage Doctors

Features:

- Verify Doctor (if someone registers as a doctor, their status will not be verified initially. Only Admin can verify the doctors)
 - Reject Doctor Verification
 - Update Doctor Status (Admin can also cancel verification of the verified doctors)
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Manage Appointments

Features:

- View All Appointments
 - Monitor Appointment Status
-

Payment Management

Features:

- View Payment Records
-

Analytics

Display:

- Doctor Performance (Rating based)
- Total Patients
- Total Doctors
- Total Appointments

*** Use Recharts

Payment System

Integrate Stripe Payment Gateway.

Patients must pay consultation fees before confirming appointments.

Payment status must be stored in database

Additional Requirements

Loading Page

Create a meaningful loading page for:

- Route loading
 - Data fetching
 - Dashboard loading
-

Error Page

Custom 404 Page.

Must Include:

- Illustration
 - Error Message
 - Back Home Button
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Other Requirements

- Toast/SweetAlert Notifications
 - Protected Routes
 - Role-Based Dashboard Access
 - Responsive Dashboard
 - Dynamic Page Titles
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UI Design Requirements

Unique Design

The design must not resemble previous assignments or module projects.

Use modern healthcare-themed color palettes.

Consistency

- Consistent typography
 - Consistent heading style
 - Consistent button design
 - Equal card sizes
 - Proper spacing and alignment
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Dashboard UI

- Full-width layout
 - Charts and analytics
 - User profile section
 - Responsive sidebar
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Responsiveness

Must support:

- Mobile Devices
 - Tablets
 - Desktop Devices
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Challenge Requirements

Challenge 1

Advanced Doctor Search

Search doctors by:

- Name
 - Specialization
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Challenge 2

Sorting Functionality

Sort doctors by:

- Consultation Fee
 - Experience
 - Highest Rating
-

Challenge 3

JWT Token Verification

Requirements:

- Protect Private APIs
- Verify Tokens in Backend
- Role-Based Authorization

Documentation required in README.

Challenge 4

Pagination

Implement pagination on:

- Find Doctors Page
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Optional Requirements

Choose any 2–5.

Option 1

Dark/Light Theme Toggle.

Option 2

Doctor Availability Calendar.

Option 3

Email Appointment Reminders.

Option 4

Layout Change Option (Table to Card / Card to Table format)

What to Submit

Admin Credentials

Admin Email:

Admin Password:

Links

Live Site Link:

GitHub Repository (Client):

GitHub Repository (Server):

Technology Suggestions

Frontend

- Next Js
- Tailwind CSS
- DaisyUI / HeroUI
- Framer Motion
- Recharts

Backend

- Node.js
- Express.js
- MongoDB

- JWT
- Stripe

Authentication

- Better Auth